

Fast-Locking Electro-Optic PLL with Harmonic Suppression and Lock-in Calibration for FMCW LiDAR

Jinhai Xiao, Ningyi Zhang, Rui Liu, Chenghao Zhang, Maliang Liu,
and Yintang Yang, *Senior Member, IEEE*

Abstract—An Electro-Optic Phase-Locked Loop (EOPLL) is introduced for application in Frequency Modulated Continuous Wave (FMCW) Light Detection And Ranging (LiDAR), aimed at mitigating the impact of laser nonlinearity, temperature variations, and voltage fluctuations. Nonlinearity in the laser can significantly affect the detection accuracy of LiDAR. The proposed EOPLL substantially improves the linearity of the laser from 90% to 99.9% within a temperature range of 10-60 °C. The proposed EOPLL exhibits an advantage over wideband PLLs by incorporating Harmonic Suppression (HS) mixing into the loop, effectively mitigating spurious tones. Additionally, a bleed current phase adjustment technique has been introduced to align the phase input-to-clock for lower harmonics. Consequently, a significant 29dB enhancement in suppressing the third harmonic of the beat signal is attained. The bandwidth of FMCW LiDAR is critical because the extend of the lock time affects the effective detection bandwidth. In response to relocking issues associated with EOPLLs, a switchable Lock-in calibration technique is proposed to reduce the locking time and increase the effective bandwidth (EBW) from 50% to 90%.

Index Terms—FMCW, EOPLL, LiDAR, Harmonic Suppression, 3D imaging.

I. INTRODUCTION

3D imaging technology finds extensive applications across diverse sectors. It plays a crucial role not only in industrial applications but also in the consumer market. The applications of 3D imaging have expanded from autonomous vehicles [1] - [3] to medical diagnostics [4][5], and from architectural design to cultural preservation. This technology has surpassed its initial use as an aviation 3D mapping tool, evolving into a key technology capable of high-precision imaging over short [6] - [10] (1 meter) to long [11] - [15] (100 meters) distances. Advanced driver-assistance systems (ADAS) and autonomous vehicles have the potential to significantly reduce pedestrian-related vehicular accidents and fatalities, decrease costs of commercial truck accidents, and alleviate traffic congestion in major cities. To achieve this, visual camera imaging and scanning LiDAR technologies have been integrated into autonomous vehicles for navigation, positioning, and object recognition. The combination of FMCW LiDAR and

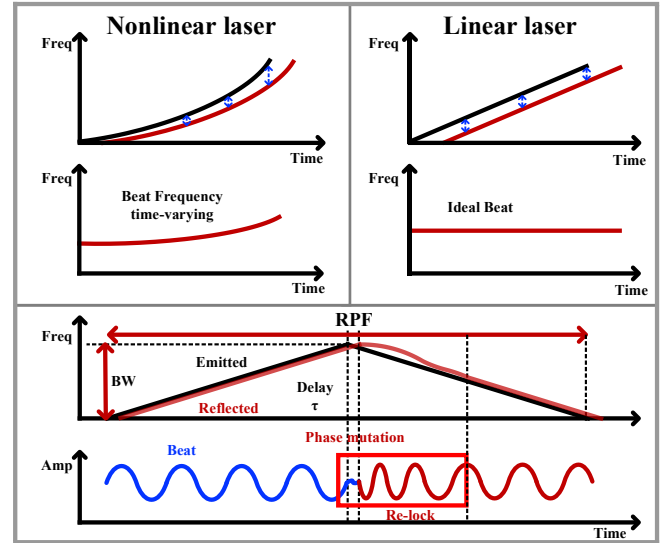


Fig. 1. The principle of FMCW LiDAR and the nonlinear principle of Laser.

Optical Phased Array (OPA) [16] is inherently suited for constructing solid-state LiDAR. Due to its high integration, accuracy, and high-speed data transmission capabilities, FMCW LiDAR has the potential to emerge as the next generation of LiDAR technology. Compared to the scanning laser LiDAR systems, EOPLL integrates more conveniently with solid-state OPA to form a solid-state FMCW LiDAR.

However, continuous frequency-modulated lasers vary the wavelength [17], rendering inaccurate distance measurements, and thus, cannot be directly used in FMCW LiDAR. To enhance the linearity of FMCW LiDAR, various EOPLL have been proposed, such as external FPGA calibration systems[18], off-chip pre-distortion calibration systems[19], and integrated PLL systems [7][10]. With its highly integrated features, solid-state EOPLL is more conducive to the development of low-cost FMCW LiDAR.

In traditional EOPLL, there is a significant trade-off between the linear frequency modulation bandwidth and the delay (τ) of the Mach-Zehnder Interferometer (MZI) used for laser linear frequency modulation. EOPLLs require a large-delay waveguide in the MZI for modulation coefficient. However, a large-delay waveguide occupies a significant portion of the silicon chip area and introduces substantial losses. This diverts most laser power to the EOPLL, posing challenges in power-constrained applications.

Jinhai Xiao, Maliang Liu, and Yintang Yang are with Shaanxi Key Lab. of Integrated Circuits and Systems, School of Microelectronics, Xidian University, Xi'an, 710071, China (e-mail: xiaojinhai@xidian.edu.cn; mliu@xidian.edu.cn; and ytyang@xidian.edu.cn)

In an EOPLL system, the delay τ_{MZI} needs to be as small as possible to reduce the packaging area. This results in a low beat frequency, yielding few beat cycles within a single repeat period, which increases the requirements for the settling time of the EOPLL (as shown in Fig. 1). The long settling time prevents the EOPLL from contributing to the effective detection bandwidth, thus deteriorating the system's output bandwidth and reducing the overall accuracy. For example, for an autonomous driving FMCW LiDAR with an output bandwidth (BW) of the laser at 4 GHz, a delay time of 1 ns, a repetition frequency (RPF) of 4 kHz, and a beat frequency of 32 kHz, there are only 4 beat tone periods, as shown in Fig. 1. Achieving settling within 4 CLK cycles is challenging because relocking is required each time the direction of the drive triangle wave is switched. Therefore, the settling time of the EOPLL system is very important.

To enhance the locking speed, reduce spurious emissions, and improve the integration of the EOPLL, a solid-state EOPLL chip was designed based on the 65nm standard CMOS process in this work. An EOPLL structure was proposed to circumvent interference from traditional laser factors such as process, voltage, and temperature (PVT). This has improved the laser's linearity from 90% to 99.9%.

Compared with the single-point locking in [10], a new polyphase filter (PPF) structure is introduced, significantly expanding the frequency range of IQ conversion and tripling the bandwidth of the single-point locking range. A new Eight-phase mixing structure was proposed to enhance the original mixer's harmonic suppression by over 29dB.

Traditional EOPLL requires relocking at the modulation triangle wave switching points. To prevent wastage of effective bandwidth due to relocking, a lock-in calibration (LIC) module was introduced, reducing the locking time from the original 50% of the repetition period to less than 10%. Additionally, voltage fluctuations in laser bias further exacerbate nonlinearity, which is mitigated by the proposed closed-loop EOPLL.

This paper discusses the proposed architecture in Section II, followed by the transfer function analysis in Section III, the circuit implementation in Section IV and measurements in Section V. Finally, conclusions are drawn in Section VI.

II. PROPOSED EOPLL WITH HSM

A. Comparison of three different architectures of the EOPLL

1) DT-EOPLL

To improve the linearity of the FMCW laser, the charge pump structure EOPLL[7], as depicted in Fig. 2(a), has been shown to effectively enhance its linearity. However, the discrete-time (DT) EOPLL requires a loop bandwidth less than ten times the reference frequency, which results in a prolonged locking time[20]. This demands that the EOPLL system have a sufficient number of beat frequencies within the repetition period.

The lock time of traditional CPPLL is:

$$T_L = \frac{k}{BW} \quad (1)$$

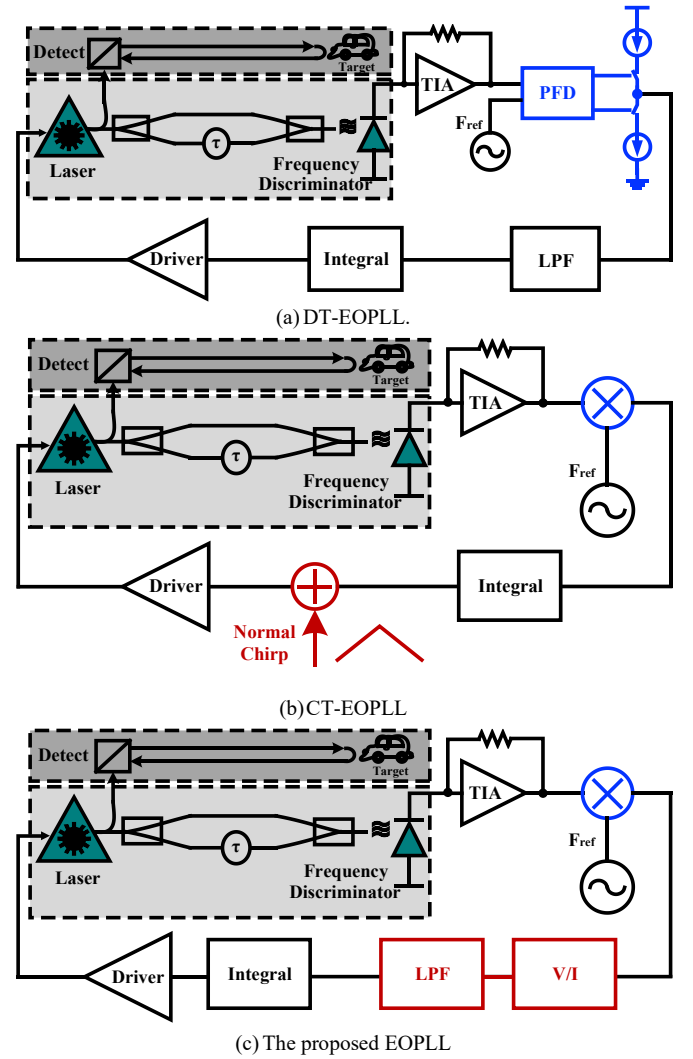


Fig. 2. (a)(b)(c) EOPLL with different structures.

Where BW represents the loop bandwidth of the PLL, and k is a factor related to the PLL phase margin. Since the charge pump operates as a discrete-time system, its bandwidth (BW) must be less than ten times the reference frequency to ensure stability. This means the EOPLL requires more than ten reference cycles to stabilize, which contradicts the need for fast locking.

2) CT-EOPLL

As shown in Fig. 2(b), replacing the digital phase detector with a continuous-time (CT) analog mixer allows for continuous correction of linear frequency modulation slope errors. This configuration enables the loop bandwidth to match the reference frequency, effectively reducing the locking time[10]. However, CT-EOPLL introduces a substantial amount of harmonics because CT-EOPLL will generate an intermodulation tone at the sum of the beat tone and the reference. This generates spurious signals at the offset of the expected beat tone in the range measurement output, necessitating the addition of extra harmonic suppression circuits. In the initial state, both positive and negative outputs of the mixer are equal. a the formation of degeneracy points and ensure the integrator can produce a triangular wave, it's essential to integrate the EOPLL with the regular chirp. This

approach requires support from an additional high-precision DAC. Compared to digital pre-distortion, this method doesn't reduce hardware requirements.

3) The proposed EOPLL

The proposed structure is shown in Fig. 2(c), which introduces a V/I conversion circuit and loop filter (LPF). This design focuses on fast locking without requiring an additional chirp signal, thereby reducing hardware requirements. The V/I converter converts the output voltage of the mixer into a current, which is integrated in LPF to produce the integrator input. This configuration not only suppresses mixer output harmonics via LPF filtering, but also stabilizes the phase relationship between the mixer's input clock and beat signal. Consequently, the mixer output voltage is maintained at an operating point with minimized harmonic distortion.

III. EOPLL TRANSFER FUNCTIONS ANALYSIS

The phase domain model of EOPLL is shown in the Fig. 3. The core module functions of it are the same as those of the traditional electrical phase-locked loop, and it is easy to extend the phase domain model of the traditional PLL to that of EOPLL.

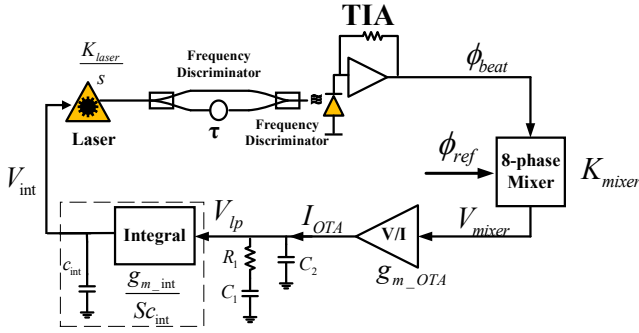


Fig. 3. The phase domain model of EOPLL

The mixer detects phase differences by generating a proportional voltage (V_{mixer}). Its transfer function is typically expressed as:

$$V_{mixer}(s) = K_{mixer}[\phi_{ref}(s) - \phi_{beat}(s)] \quad (2)$$

Where K_{mixer} denotes the Mixer gain, $\phi_{ref}(s) - \phi_{beat}(s)$ is the phase difference between the input reference signal and the feedback beat signal.

The output voltage of the mixer is converted into current (I_{OTA}) input into the loop filter by a transconductance amplifier. The mixer and transconductance amplifier are equivalent to phase discriminator and charge pump, and the overall phase discrimination gain is:

$$K_{pd} = \frac{I_{OTA}(s)}{\phi_{ref}(s) - \phi_{beat}(s)} = K_{mixer}g_{m_OTA} \quad (3)$$

Where g_{m_OTA} is transconductance amplifier conversion gain.

The impedance transfer function of the conventional second-order loop filter is:

$$F_{lp}(s) = \frac{V_{lp}(s)}{I_{OTA}(s)} = \frac{1 + sR_1C_1}{s^2R_1C_1C_2 + s(C_1 + C_2)} \quad (4)$$

Where V_{lp} is the output voltage of the loop filter.

Semiconductor laser frequency modulation is achieved by bias voltage modulation in the gain section. Thus, the integrator, laser, and delay line are collectively modeled as a voltage-controlled oscillator (VCO).

The integrator performs integration by charging a capacitor through a transconductance amplifier. If the transconductance of the transconductance operational amplifier is g_{m_int} and the load capacitance is C_{int} , the output voltage of the integrator is:

$$V_{int}(s) = \frac{g_{m_int}V_{lp}(s)}{sC_{int}} \quad (5)$$

The relationship between the phase of laser output (ϕ_{laser}) and the input voltage is:

$$\phi_{laser}(t) = \omega_0 t + \int K_{laser} V_{int}(t) dt \quad (6)$$

Where ω_0 is the angular frequency corresponding to the bias voltage, K_{laser} is the voltage-frequency conversion gain of the laser.

The laser output signal is divided into two channels, one with τ delay, the other without delay, the photodetector will be coherent to get the beat signal.

$$\begin{aligned} \phi_{beat}(t) &= \phi_{laser}(t + \tau) - \phi_{laser}(t) \\ &= \omega_0 \tau + \int_0^{t+\tau} K_{laser} V_{int}(t) dt - \int_0^t K_{laser} V_{int}(t) dt \end{aligned} \quad (7)$$

Applying the Laplace transform to Equation (7) yields:

$$\phi_{beat}(s) = \frac{1}{s} \cdot K_{laser} V_{int}(s) (e^{s\tau} - 1) \approx K_{laser} V_{int}(s) \tau \quad (8)$$

The equivalent VCO transfer function is therefore:

$$\frac{\phi_{beat}}{V_{lp}(s)} = \frac{K_{laser} g_{m_int} \tau}{sC_{int}} = \frac{K_{vco}}{s} \quad (9)$$

Finally, the open-loop transfer function of the EOPLL enabled is:

$$OLTF(s) = K_{pd} \cdot \frac{K_{vco}}{s} \cdot F_{lp}(s) = \frac{K_{pd} K_{vco} (1 + sR_1C_1)}{s^2 [sR_1C_1C_2 + (C_1 + C_2)]} \quad (10)$$

IV. CIRCUIT IMPLEMENTATION

This section details the circuit implementation, including the EOPLL's overall architecture and descriptions of its modules.

A. EOPLL Design

The EOPLL proposed is illustrated in Fig. 4. The EOPLL system consists of a transimpedance amplifier (TIA), a single-ended to differential amplifier (SDA), a PPF, a 4-8 phase converter (FEPC), a harmonic suppression mixer (HSM), a CLK generator, a voltage/current converter (VIC), switches, integrators, and drivers. The VIC output voltage is integrated by an integrator to generate a triangular wave, and the laser is modulated by a laser driver. The delay time produced by the MZI is τ_{MZI} . Here, the delay corresponds to the time it takes for the laser to travel from emission to the object surface and back. The beat signal is generated by coherent mixing of the delayed

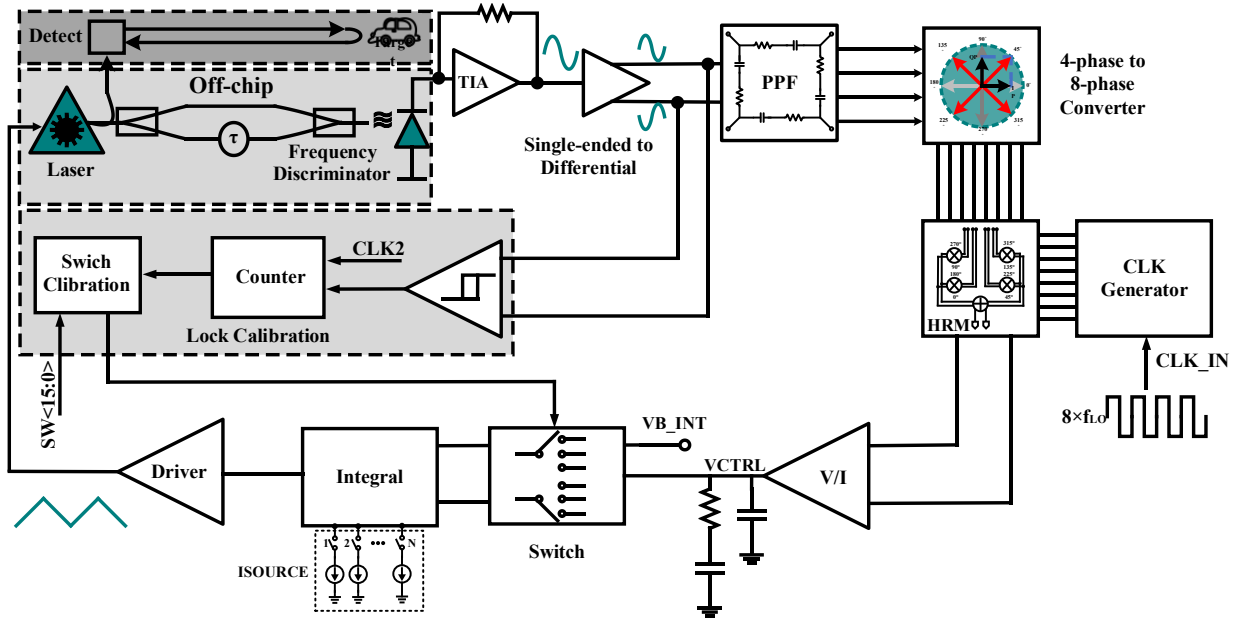


Fig. 4. The proposed EOPLL.

and undelayed optical paths. The beat signal is detected by a photodiode (PD) and is converted into a differential voltage through the TIA and SDA. This signal undergoes three-stage PPF to generate 0° , 90° , 180° , and 270° signals, which are further converted into 0° , 45° , 90° , 135° , 180° , 225° , 270° , and 315° signals by the FEPC. Then, the eight-phase signals are each mixed with the eight-phase CLK. The output of the mixer, after V/I conversion and loop filtering, is connected to the switching module. The switch generates a triangular wave by alternating between two voltage poles.

The mixer's output directly serves as the input for the integrator. During startup, identical DC voltages from the mixer may prevent triangular wave generation, causing lock failure. To mitigate this issue, a V/I converter and a loop filter were added. This loop filter not only allows for easy adjustment of the loop bandwidth according to different lasers but also filters out the high-frequency harmonics of the mixer.

B. Harmonic Suppression Mixer

In CT-EOPLL designs, beat signal harmonics originate primarily from mixer output harmonics. These harmonics from the mixer directly couple into the integrator, subsequently influencing the spurious content of the laser signal. Also, unlike conventional harmonic rejection mixers, the mixer outputs used in EOPLLs operate near DC.

The expressions defining the mixer input (sine signal) and LO (square wave signal) are as follows:

Quadrature Mixer output:

$$f_{out}(t) = \frac{4}{\pi} \left[\cos(\omega_0 t - \omega_1 t) - \frac{1}{3} \cos(3\omega_0 t + \omega_1 t) + \frac{1}{5} \cos(5\omega_0 t - \omega_1 t) - \dots \right] \quad (11)$$

Eight-phase mixer output:

$$f_{out}(t) = -\frac{8}{\pi} \left[\cos(\omega_0 t - \omega_1 t) - \frac{1}{7} \cos(7\omega_0 t + \omega_1 t) \right] \quad (12)$$

Upon examining the outputs of the ideal quadrature mixer and the eight-phase mixer, the quadrature mixer outputs components $\omega_0 - \omega_1$, $3\omega_0 + \omega_1$, $5\omega_0 - \omega_1$, $7\omega_0 + \omega_1$, and so on. In contrast, the eight-phase mixer only outputs components $\omega_0 - \omega_1$ and $7\omega_0 + \omega_1$. Compared with quadrature mixers, eight-phase mixers push low-frequency harmonics to high frequencies through multi-phase clocks, effectively suppressing $3\omega_0 + \omega_1$, $5\omega_0 - \omega_1$ and other harmonics.

However, amplitude and phase mismatches generate multiple harmonics, which impose stringent requirements on the input amplitude matching for the mixer. Given the multitude of phases, when both phase and amplitude are simultaneously considered, the equation becomes particularly complex and difficult to simplify. Furthermore, for the sake of simplification, only the third harmonic is considered. When there's a phase discrepancy between IQ signal, the formula for the third harmonic reads as follows:

$$\begin{aligned} HRM_{3phase} &= -\frac{2\sqrt{2}}{3\pi} \left(\cos(3\omega_0 t - \omega_1 t - \frac{\pi}{4}) \right. \\ &\quad \left. - \cos(3\omega_0 t - \omega_1 t - \frac{\pi}{4} - \theta) \right) \\ &= \frac{2\sqrt{2}}{3\pi} \sqrt{2 - 2\cos(\theta)} \left(\cos(3\omega_0 t - \omega_1 t - \frac{\pi}{4}) \right) \end{aligned} \quad (13)$$

When considering the amplitude gain error, the formula for the third harmonic reads as follows:

$$\begin{aligned} HRM_{3gain} &= f(\omega_0 t - \frac{\pi}{2}) \Delta \sin(\omega_1 t - \frac{\pi}{2}) \\ &\quad + f(\omega_0 t - \frac{3\pi}{4}) \Delta \sin(\omega_1 t - \frac{3\pi}{4}) \\ &= \frac{2\sqrt{2}}{3\pi} \Delta \cos(3\omega_0 t - \omega_1 t - \frac{\pi}{4}) \end{aligned} \quad (14)$$

Where θ is the phase mismatch and Δ is the gain mismatch.

In[21], the intermediate stage mixer's current is set to be $\sqrt{2}$ times its initial value. However, significant amplitude mismatches persist. This configuration assumes that the beat signal is an ideal sine wave, but elements like MZI, TIA, and SDA invariably introduce distortions. As shown in Fig. 5, when the mixer's input signal experiences distortion, the $\sqrt{2}$ multiplication doesn't effectively align the mixer's sampling points, compromising its harmonic suppression capability. To enhance harmonic suppression, the four input signals are converted to an eight-phase pattern. Fig. 5 illustrates that the mixer's sampling point voltage of the eight-phase input remains more consistent compared to that of the four-phase counterpart. This stability is maintained even if the input signal deviates from an ideal sine wave, ensuring equalized sampling point voltages. To mitigate potential mismatches and distortions introduced by these components, careful design considerations are necessary, and compensation circuits may be employed. It's crucial to monitor the actual operational conditions of the mixer and its preceding stages to ensure consistent performance across the desired frequency range. The eight-phase input offers an optimal trade-off between performance and phase error control. Increasing the phase number further would cause phase inaccuracy, while fewer phases would lead to higher spurs.

For EOPLL, once locked, the mixer outputs a DC signal, with harmonics originating from harmonic components superimposed on the DC level. As shown in Fig. 6(a)(b), the conventional HSM is contrasted with the HSM mixer introduced in this paper. The proposed mixer is composed of two sets of IQ mixers with a 45-degree phase difference

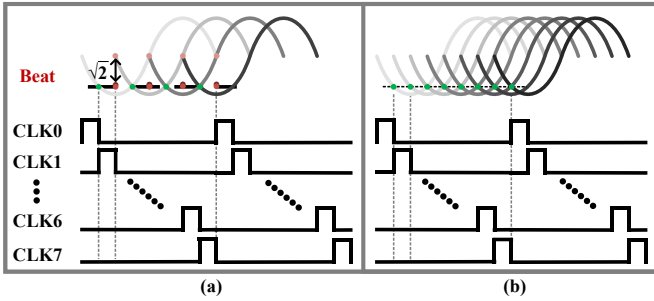


Fig. 5. Comparison of sampling effect between four-phase input and eight-phase input HSM.

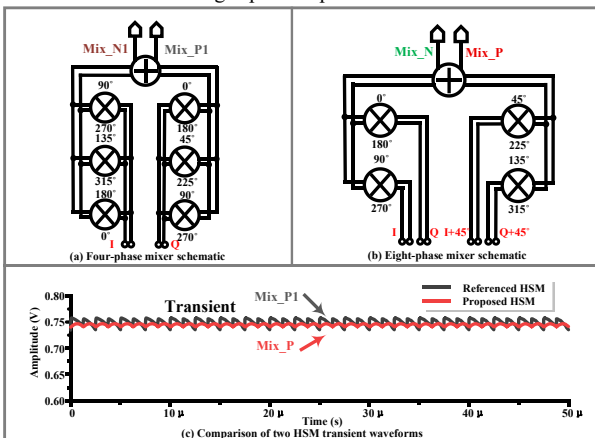


Fig. 6. Comparison of Four-phase HSM and the proposed Eight-phase HSM.

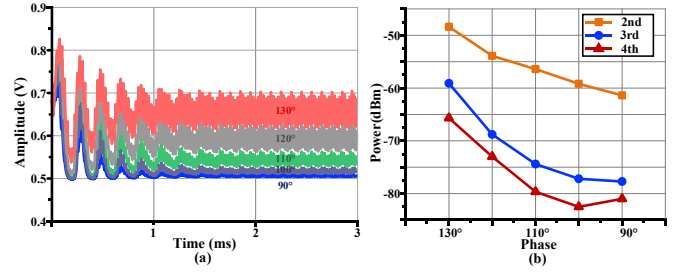


Fig. 7. HSM (a) output amplitude and (b) its harmonic power for different input-to-clock phase differences (Simulated results).

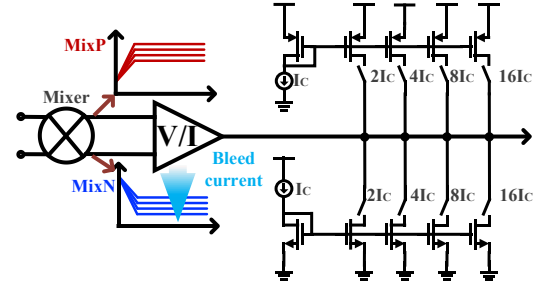


Fig. 8. Schematic of bleed current changing mixer output.

between them. When the proposed mixer is operated with an input frequency equal to the clock frequency, the output is illustrated in Fig. 6(c), alongside the conventional HSM mixer. With the same input-to-clock phase difference, the equidistant and equal amplitude mixing sampling provide a balanced advantage, enhancing the harmonic suppression performance of the proposed mixer at each harmonic point.

The harmonics close to the DC output are influenced by the phase relationship between the clock and the input, as depicted in Fig. 7. From the figure, it can be observed that when the phase difference is at 90 degrees, the CLK sampling points align with the peak of the input sine wave, resulting in minimal output harmonics. As the phase difference increases, the sampling points deviate from the peak of the sine wave, leading to a gradual increase in the mixer's output harmonics. It's crucial to optimize the design to achieve harmonic suppression, as harmonics can interfere with signal quality and system performance. Utilizing appropriate biasing and considering the phase relationship between signals can assist in achieving better harmonic control.

To suppress the mixer's output harmonics as much as possible, the output phase should be consistently maintained at a point where harmonics are minimized. Compared to the mixer structure in [10], where the output varies with the drive laser, the VIC structure can fix the mixer sampling points at a stable position, ensuring a consistent input-clock phase difference under different non-linear conditions. To form a voltage offset at the mixer's output, a bleed current can be introduced at the VIC's output point. This effectively positions the mixer's sampling points near the peak of the beat signal, thereby minimizing output harmonics. The bleed current structure is illustrated in Fig. 8, where a current mirror at the VIC output controls the bleed current, in turn controlling the mixer's output voltage offset. However, the magnitude of the bleed current should be a compromise with the VIC output current to avoid destabilizing the loop due to the bleed current.

In order to realize eight-phase mixing, it is necessary to convert the beat signal into an eight-phase signal. The eight-phase beat signal generation is shown in Fig. 9, consisting of the PPF circuit and vector sum circuit. The differential signal is converted to a quad-phase signal through the PPF. Then, the quad-phase signal undergoes a vector summation where signals with a 90-degree phase difference are added together. The original quad-phase signals are added to themselves, ultimately producing an eight-phase signal with a 45-degree phase difference. The phase error between different PPF outputs and the ideal IQ signal is shown as Fig. 10. The three-stage PPF achieves quadrature signals with $< \pm 1^\circ$ within the frequency range of 100 - 500 KHz. By adopting multi-stage cascading, its bandwidth can be wider than that of a single-stage PPF, making it suitable for more lasers and τ_{MZI} . The vector addition circuit combines quadrature signals to generate eight phases.

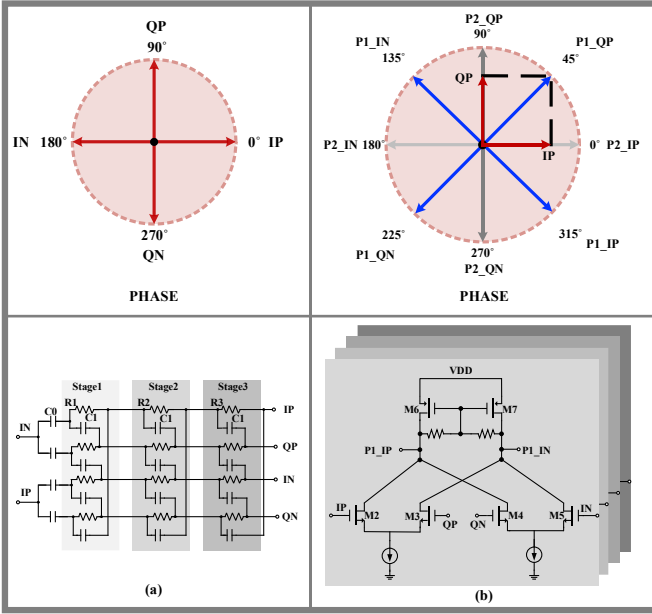


Fig. 9 (a) Schematic and phase of PPF, (b) Schematic of vector addition circuit and corresponding phase.

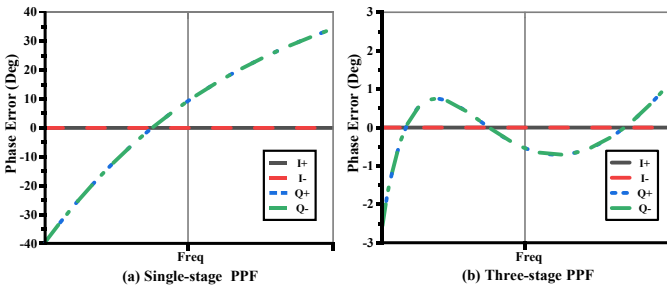


Fig. 10. Phase mismatch of (a) Single-stage PPF and (b) Three-stage PPF output (Simulated results).

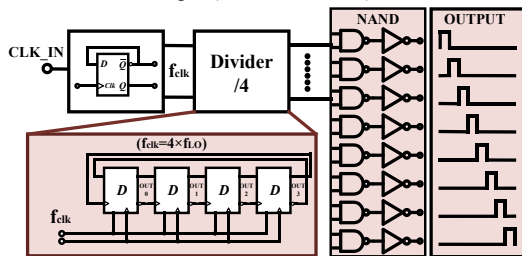


Fig. 11. Clock generation circuit.

The clock generation circuit is depicted in Fig. 11. The circuit produces eight clock signals with a 12.5% duty cycle, comprised of a divide-by-four circuit using D flip-flops and a pulse generation circuit. To generate the eight-phase signal, the input clock should be four times the frequency of the reference clock. After division, the signals undergo a logical NAND operation, ultimately resulting in eight non-overlapping clock signals. These non-overlapping clocks prevent undesirable factors, such as harmonic leakage, that might arise when different signals simultaneously conduct during the mixing process.

This design takes advantage of multi-phase techniques to enhance signal processing and manipulation capabilities, allowing for a more versatile and effective system. Multi-phase techniques with vector addition enhance harmonic rejection and signal fidelity.

C. Lock-in Calibration

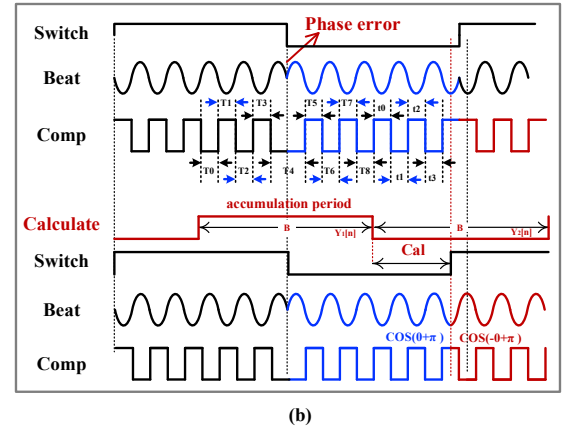
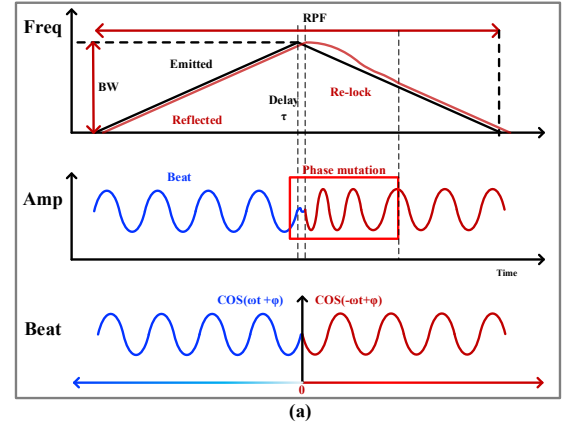


Fig. 12. (a) Mechanism of phase mutation of beat frequency signal. (b) Calibration process.

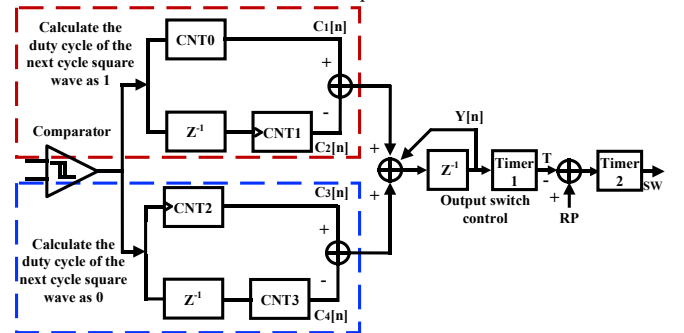


Fig. 13. Calibration algorithm block diagram.

The phase-locked state of the phase-locked loop (PLL) must be maintained with continuous phase. As shown in Fig. 12(a), the FMCW LiDAR will form a phase mutation at the transition point of the repetitive triangle wave. This will cause the EOPLL to re-lock. Not only will this pose problem in signal processing, but it will also narrow the effective bandwidth of the FMCW signal output. Therefore, it's essential to allow the PLL to maintain continuous locking during the modulation triangle wave transition.

At the triangle wave transition point, assuming the time is 0, the beat signals on both sides are represented as $\cos(\omega t + \varphi)$ and $\cos(\omega t - \varphi)$. It's clear that only when switching at the peak of the beat signal, can the beat signal remain continuous. To address this, a lock calibration technique is proposed as shown in Fig. 12(b). The block diagram of LIC algorithm is shown in Fig. 13. First, the comparator converts the beat signal into a pulse signal. Then, the duty cycle of the beat signal is calculated using a counter. By comparing the duty cycles before and after each switch, and then iterating through an algorithm, the switching point of the beat signal is moved to an approximately continuous beat location. This eliminates the need for re-locking. In the calibration system, the calculation for the duty cycle error involves:

$$x[n] = \beta(c_1[n] - c_2[n] + c_3[n] - c_4[n]) \quad (15)$$

where $c_1[n]$ and $c_2[n]$ represent the count duration of the previous and subsequent pulses with value 1, respectively. $c_3[n]$ and $c_4[n]$ represent the count duration of the previous and next pulse with a value of 0. β is the cumulative scaling factor.

As shown in Fig. 12(b), to accumulate the phase error caused by the directional switch, phase comparison starts from a quarter of the RPF period and accumulates every RPF period. The repetition error calibration occurs at one quarter and three quarters of the RPF period. In this manner, although the length of the RPF period might vary during calibration, it becomes the accurate set value after calibration. For example, after calibration, the accumulated error $Y_1[n]$ is equal to the accumulated error $Y_2[n]$ of the next cycle. The final ST and T are equal, and the repetition time is also equal. The resulting output from the calibration algorithm is:

$$Y[n] = \sum_{k=0}^{n-1} x[k] \quad (16)$$

$$ST = RP - T \quad (17)$$

Here, RP signifies the duration of the repetitive cycle, T is the delay time of phase error accumulation $Y[n]$.

V. EOPLL MEASUREMENTS

The EOPLL test system is shown in Fig. 14(a), the EOPLL chip photo and test PCB are shown in Fig. 14(b), and the test environment is shown in Fig. 14(c). The system includes EOPLL chip, laser, MZI, APD, FPGA and oscilloscope. The FPGA is used to control parameters such as the repetition frequency period of the chip, and the oscilloscope is used to record experimental data. The EOPLL system utilizes a 1550 nm DFB laser. The test system and chip photos are shown in

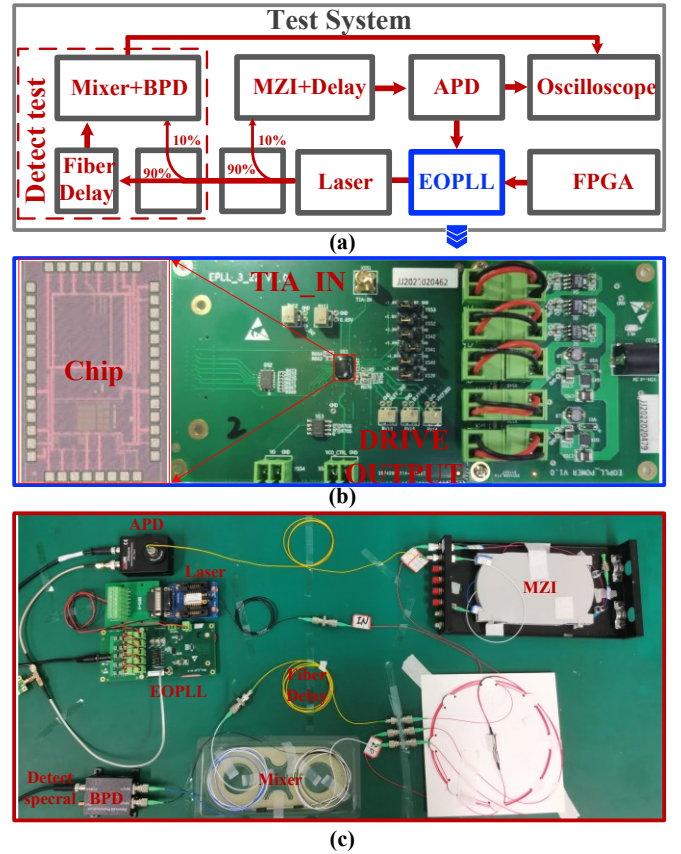


Fig. 14. (a) EOPLL test system, (b) chip photo and test PCB, and (c) test environment.

Fig. 14(b). The chip has a power consumption of 68 mw, excluding the driver. The system features a 5 GHz laser bandwidth, a delay of 2 ns, a 5 KHz RPF, a 100 KHz beat, and a reference clock frequency of 8×100 KHz.

For different configurations of the LiDAR system, including open-loop and various closed-loop setups, the frequency domain results of the EOPLL TIA beat output can be seen in Fig. 15. It is evident that the beat frequency is highly unstable and fluctuates over time, resulting in a large frequency bandwidth for the beat signal, as shown in Fig. 15(a). In a closed-loop, the beat signal becomes highly stable with the spectral peak remaining consistent, as shown in Fig. 15(b). As shown in Fig. 15(c), when the proposed HSM is integrated, the beat frequency is further refined. It can be seen that the second harmonic is suppressed by more than 44 dB and the third harmonic is suppressed by more than 58 dB. Fig. 16 shows the frequency error comparison before and after EOPLL locking, it can be seen that EOPLL reduces the frequency error of the laser from 200M to the KHz level. The results of using optical fibers to simulate detection distances are presented in Fig. 17. Beat signals from fibers with lengths of 1m, 2m, 4m, and 8m were tested. As the fiber length increases, the spectrum of the beat signal broadens. According to [22], to detect signals over 100 m, the laser's linewidth should be narrower than 100 Hz. To further validate robustness against voltage fluctuations, controlled bias variations (± 0.05 V) were applied to the laser. As shown in Fig. 18, the EOPLL maintains consistent beat signal linearity under

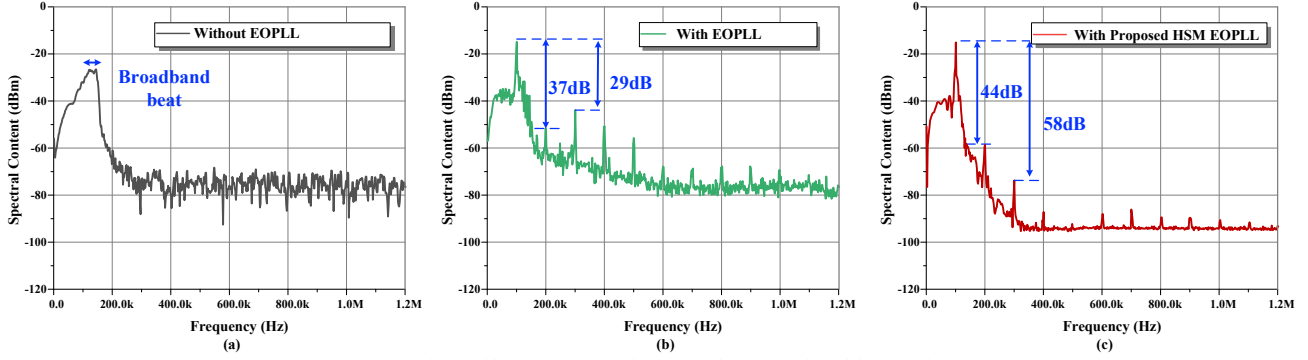


Fig. 15. Comparison of beat spectrum of (a) open-loop, (a) closed-loop and (a) HSM.

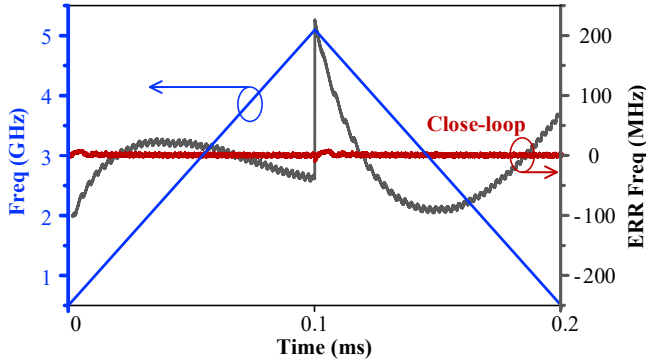


Fig. 16. Comparison of beat frequency error before and after locking.

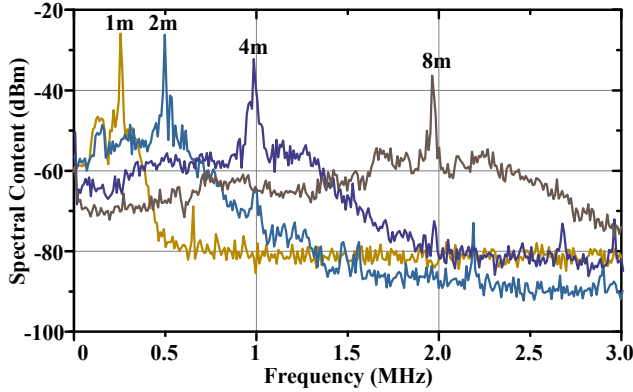


Fig. 17. Spectral Results of Different Lengths of Optical Fiber.

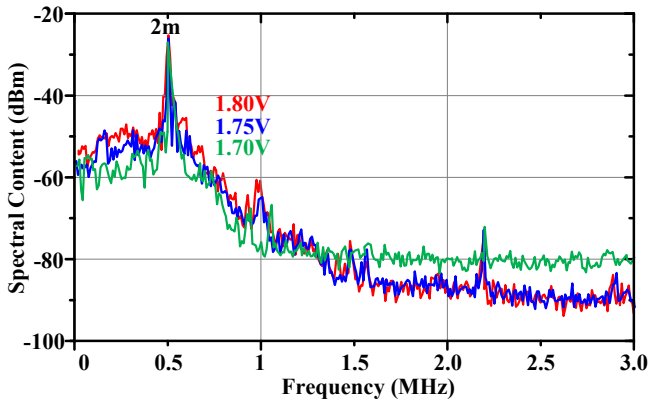


Fig. 18. EOPLL beat spectra under laser bias variations. 1.7 V, 1.75 V, and 1.8 V bias conditions, confirming its ability to compensate for voltage-induced nonlinearities.

To avoid re-locking, we introduced a LIC that switches only at the peak values of the captured sinusoidal wave, effectively reducing the re-locking time of the EOPLL. Fig.

19(a) depicts the locked state before calibration, while Fig. 19(b) illustrates the locked state after calibration. Fig. 19(c) represents the locked state with a narrower bandwidth, demonstrating that $BW \cdot \tau_{MZI}$ can reach a minimum of 5. Results show that the lock-in calibration reduces the locking time from 50% (5 beats frequency period) to less than 10% (1 beat period) of the repetition period.

Fig. 20 illustrates the comparison of laser linearity before and after the introduction of EOPLL. It can be observed that within the operating temperature range of the laser, the linearity without EOPLL is approximately 90% and deteriorates with increasing temperature. However, after introducing EOPLL, the laser linearity exceeds 99.9% and remains insensitive to temperature variations. TABLE I summarizes the performance of the proposed system and compares it with existing techniques. The proposed technique achieves excellent spurious suppression, effective bandwidth ratio, and improved integration ($BW \cdot \tau_{MZI}$).

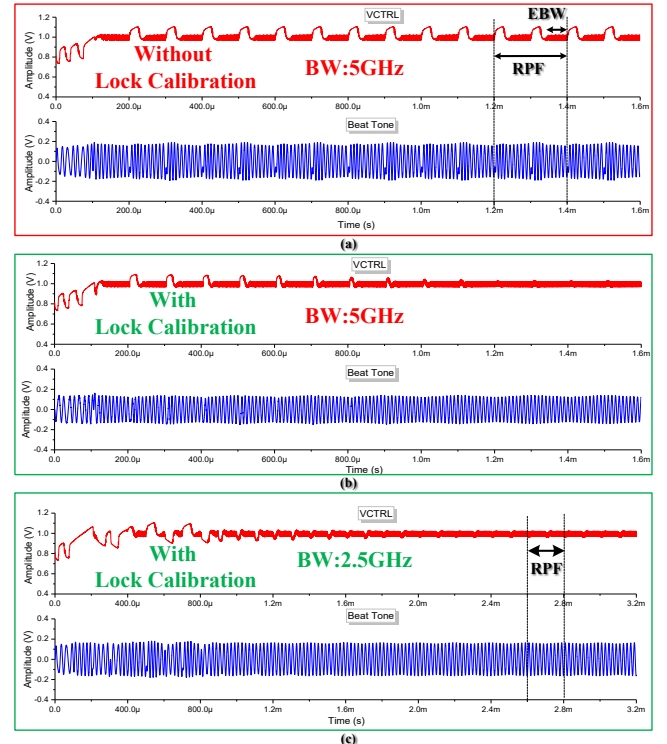


Fig. 19. Comparison before and after lock calibration (Linear frequency chirps correlate with VCTRL symmetry).

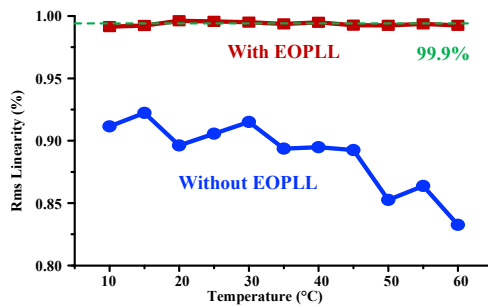


Fig. 20. Comparison of linearity at different temperatures without and with EOPLL.

TABLE I PERFORMANCE COMPARISON TO OTHER EOPLL TECHNIQUES.

	[23]	[24]	[7]	[10]	This Work
Architecture	CT-EOPLL	CT-EOPLL	DT-EOPLL	CT-EOPLL	CT-EOPLL
Technology, nm	Discret	Discret	180	65	65
Wavelength, nm	1539	1539	1550	1546	1550
Chirp BW, GHz	16.67	100	122	16	5
BW* τ_{MZI} , Hz.s	10	2860	40.26	10	5
Spur Suppression	17dB	N/A	N/A	25dB	29dB
EBW/BW	N/A	N/A	N/A	N/A	90%

The performance of the proposed architecture is summarized and compared to prior art in TABLE I. Owing to the enhancements in the CT-EOPLL structure, superior spur suppression performance has been achieved, paving the way for potential long-distance detection in the future. The introduced LIC results in a reduced BW* τ_{MZI} (integrability) and achieves a higher EBW.

VI. CONCLUSION

A fast-locking continuous-time EOPLL is proposed to enhance FMCW LiDAR performance. By introducing an eight-phase harmonic suppression mixer and a lock-in calibration technique, the system achieves >29 dB harmonic suppression and reduces locking time from 50% to <10% of the repetition period. The laser linearity improves from 90% to 99.9% and remains stable across 10–60 °C. Fabricated in 65 nm CMOS, the design offers high integration, low power (68 mW), and strong robustness to bias and temperature variations, making it well-suited for solid-state LiDAR applications.

ACKNOWLEDGEMENT

This work was supported in part by the Fundamental Research Funds for the Central Universities under Grant KYFZ25008, in part by the Natural Science Foundation of China under Grant 8091B02042301, in part by Shaanxi Provincial Key Research and Development Program under Grant 2024CY2GJHX34, in part by National Science and Technology Major Project under Grant 2024ZD0302600, and

in part by the innovation Fund of Xidian University.

REFERENCES

- [1] V. Winkler, "Range Doppler detection for automotive FMCW radars," in *Proc. Eur. Radar Conf. (EuRAD)*, pp. 166–169, 2007.
- [2] L. de Paula Veronese et al., "Evaluating the Limits of a LiDAR for an Autonomous Driving Localization," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 3, pp. 1449–1458, March 2021.
- [3] T. -N. Luo, C. -H. E. Wu and Y. -J. E. Chen, "A 77-GHz CMOS Automotive Radar Transceiver With Anti-Interference Function," *IEEE Trans. Circuits Syst. I, Reg. Papers.*, vol. 60, no. 12, pp. 3247–3255, Dec. 2013.
- [4] M. Alizadeh, G. Shaker, J. C. M. D. Almeida, P. P. Morita and S. Safavi-Naeini, "Remote Monitoring of Human Vital Signs Using mm-Wave FMCW Radar," *IEEE Access*, vol. 7, pp. 54958–54968, May 2019.
- [5] W. -C. Su, M. -C. Tang, R. E. Arif, T. -S. Horng and F. -K. Wang, "Stepped-Frequency Continuous-Wave Radar With Self-Injection-Locking Technology for Monitoring Multiple Human Vital Signs," *IEEE Trans. Microw. Theory Techn.*, vol. 67, no. 12, pp. 5396–5405, Dec. 2019.
- [6] N. Satyan, A. Vasilyev, G. Rakuljic, V. Leyva, and A. Yariv, "Precise control of broadband frequency chirps using optoelectronic feedback," *Opt. Exp.*, vol. 17, no. 18, pp. 15991–15999, 2009.
- [7] B. Behroozpour et al., "Electronic-photonics integrated circuit for 3D microimaging," *IEEE J. Solid-State Circuits*, vol. 52, no. 1, pp. 161–172, Jan. 2017.
- [8] J. Sharma, S. Ahasan, C. T. Phare, M. Lipson, and H. Krishnaswamy, "Continuous-time electro-optic PLL with decimated optical delay/loss and spur cancellation for LiDAR," in *Proc. Conf. Lasers Electro Opt.*, 2018, pp. 1–2.
- [9] R. K. Ula, Y. Noguchi and K. Iiyama, "Three-Dimensional Object Profiling Using Highly Accurate FMCW Optical Ranging System," *J. Lightw. Technol.*, vol. 37, no. 15, pp. 3826–3833, 1 Aug. 2019.
- [10] A. Binaie, S. Ahasan and H. Krishnaswamy, "A Spurless and Wideband Continuous-Time Electro-Optical Phase Locked Loop (CT-EOPLL) for High Performance LiDAR," *IEEE Open J. of the Solid-State Circuits Society*, vol. 1, pp. 235–246, 2021.
- [11] K. Sayyah et al., "Fully Integrated FMCW LiDAR Optical Engine on a Single Silicon Chip," *J. Lightw. Technol.*, vol. 40, no. 9, pp. 2763–2772, 1 May 2022.
- [12] Y. Wu, L. Deng, K. Yang and W. Liang, "Narrow Linewidth External Cavity Laser Capable of High Repetition Frequency Tuning for FMCW LiDAR," *IEEE Photon. Technol. Lett.*, vol. 34, no. 21, pp. 1123–1126, 1 Nov. 2022.
- [13] M. Pu et al., "Dual-Heterodyne Mixing Based Phase Noise Cancellation for Long Distance Dual-Wavelength FMCW Lidar," in *Proc. Opt. Fiber Commun. Conf.*, San Diego, CA, USA, 2020, pp. 1–3.
- [14] M. Liu, H. Liu, X. Li and Z. Zhu, "A 60-m Range 6.16-mW Laser-Power Linear-Mode LiDAR System With Multiplex ADC/TDC in 65-nm CMOS," *IEEE Trans. Circuits Syst. I, Reg. Papers.*, vol. 67, no. 3, pp. 753–764, March 2020.
- [15] C. Niclass, M. Soga, H. Matsubara, M. Ogawa and M. Kagami, "A 0.18- μ m CMOS SoC for a 100-m-Range 10-Frame/s 200 $\times$ 96-Pixel Time-of-Flight Depth Sensor," *IEEE J. Solid-State Circuits*, vol. 49, no. 1, pp. 315–330, Jan. 2014.
- [16] Y. Misugi, H. Okayama and T. Kita, "Low Power Consumption 2D Beam Scanner Integrated with Wavelength Tunable Laser Diode," in *Proc. Eur. Conf. Opt. Commun.*, Basel, Switzerland, 2022, pp. 1–4.
- [17] C. Liu, L. Lu, Y. Guo, X. Li, J. Chen and L. Zhou, "Hybrid Integrated Frequency-Modulated Continuous-Wave Laser With Synchronous Tuning," *J. Lightw. Technol.*, vol. 40, no. 16, pp. 5636–5645, 15 Aug. 2022.
- [18] M. Hauser and M. Hofbauer, "FPGA-Based EO-PLL With Repetitive Control for Highly Linear Laser Frequency Tuning in FMCW LiDAR Applications," *IEEE Photon. J.*, vol. 14, no. 1, pp. 1–8, Feb. 2022.
- [19] X. Zhang, J. Pouls, and M. C. Wu, "Laser frequency sweep linearization by iterative learning pre-distortion for FMCW LiDAR," *Opt. Exp.*, vol. 27, no. 7, pp. 9965–9974, 2019.
- [20] L. Kong and B. Razavi, "A 2.4GHz 4mW inductorless RF synthesizer," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers.*, San Francisco, CA, USA, 2015, pp. 1–3.

- [21] J. A. Weldon et al., "A 1.75-GHz highly integrated narrow-band CMOS transmitter with harmonic-rejection mixers," *IEEE J. Solid-State Circuits*, vol. 36, no. 12, pp. 2003-2015, Dec. 2001.
- [22] P. A. Morton et al., "Integrated Coherent Tunable Laser (ICTL) With Ultra-Wideband Wavelength Tuning and Sub-100 Hz Lorentzian Linewidth," *J. Lightw. Technol.*, vol. 40, no. 6, pp. 1802-1809, 15 March, 2022.
- [23] J. Sharma, S. Ahasan, C. T. Phare, M. Lipson, and H. Krishnaswamy, "Continuous-time electro-optic PLL with decimated optical delay/loss and spur cancellation for LIDAR," in *Proc. Conf. Lasers Electro Opt.*, 2018, pp. 1–2.
- [24] N. Satyan, A. Vasilyev, G. Rakuljic, V. Leyva, and A. Yariv, "Precise control of broadband frequency chirps using optoelectronic feedback," *Opt. Exp.*, vol. 17, no. 18, pp. 15991–15999, 2009.